



East West University
Department of Computer Science and Engineering
Course Outline

Course: CSE246 Algorithms (Sections: All)

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1.5	4.5
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	3 Hours/Week for 13 weeks	6 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

Prerequisite

CSE103 Structured Programming
CSE207 Data Structured

Instructor Information

Instructor: K. M. Safin Kamal
Office: AB1-303
E-mail: safin.kamal@ewubd.edu

Class Routine and Office Hour

Will be provided Separately

Course Objective

This course equips students with the knowledge and skills to analyze, design, and implement efficient algorithms using fundamental techniques such as divide-and-conquer, greedy, dynamic programming, and graph-based methods. Students will also develop problem-solving abilities applicable to unseen and real-world computational problems, forming a foundation for advanced courses in Artificial Intelligence, Machine Learning, Computer Networks, and Compiler Design.

Knowledge Profile

K3: Theory-based engineering fundamentals
K4: Forefront engineering specialist knowledge for practice
K5: Engineering design

Learning Domains

Cognitive - C2: Understanding, C3: Applying, C4: Analyzing, C5: Evaluating
Psychomotor – P2: Manipulation, P3: Precision

Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required.

EP2: Range of conflicting requirements.

EP3: Depth of analysis required.

EP4: Familiarity of issues

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course, students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	Understand and analyze the efficiency and accuracy of algorithms using basic searching and sorting algorithms, asymptotic notation, complexity analysis, and formal representations.	PO1	C3	K3	EP1
CO2	Solve and analyze computational problems by applying fundamental algorithmic paradigms such as divide-and-conquer, greedy and dynamic programming techniques.	PO1	C3, C4	K3	EP1
CO3	Apply appropriate algorithmic problem-solving techniques (graph, geometric, backtracking, branch and bound, string matching, etc.) to address unseen or real-world computational problems.	PO2	C3, C4	K3	EP1
CO4	Implement and evaluate algorithms as executable programs using suitable programming language.	PO2	C3, C4, P2, P3, A2	K3, K4	EP2, EP3, EP4

Course Topics, Teaching-Learning Methods, and Assessment Scheme

Lecture	Detailed Topics	Course Topics	Teaching-Learning Method	CO	Class Test 1	Class Test 2	Class Test 3	Mid Exam		Final Exam		PO								
					C3	C3	C3	C3	C4	C3	C4	PO1	PO2							
L-1	Algorithmic efficiency and accuracy, common paradigms, brute-force approach, pseudocode	Basic Searching and Sorting Algorithms	Lecture, In/Out Class Discussion	CO1	5			5				10								
L-2	Linear and binary search, Insertion sort and its complexity analysis																			
L-3	Bubble sort, Selection sort and their analysis																			
L-4	Time and space complexity, Asymptotic Notations, running time functions and growth rate comparison	Complexity Analysis																		
L-5	Recurrence relations, iteration method, master theorem, recursion tree, substitution method																			
L-4	Divide-and-conquer strategy, Merge sort	Common algorithmic paradigms, like Divide-and-conquer, Greedy, dynamic programming		CO2				10	10		25									
L-5	Quick sort, and its comparison with merge sort																			
L-8	Greedy paradigm, coin change, fractional knapsack, job sequencing problems																			
L-9	Activity selection, Huffman coding																			
L-10	DP vs greedy vs divide-and-conquer, tabulation and memoization, 0/1 knapsack problem																			
L-11	Coin change variants, LCS, LIS																			
L-12	Matrix chain multiplication, selected DP problems (rod cutting, rock climbing etc.)																			
L-13	MID Exam																			
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Lecture	Detailed Topics	Course Topics	Teaching-Learning Method	CO	Class Test 1	Class Test 2	Class Test 3	Mid Exam		Final Exam		PO											
					C3	C3	C3	C3	C4	C3	C4	PO1	PO2										
L-14	Graph terminology, representations, applications	Graph related algorithms	Lecture, In/Out Class Discussion	CO3			5			10	5		20										
L-15	MST properties, Prim’s and Kruskal’s algorithm																						
L-16	Dijkstra’s algorithm, Bellman–Ford algorithm																						
L-17	Floyd–Warshall algorithm, Johnson’s algorithm																						
L-18	Flow and residual networks, augmenting paths, max-flow min-cut theorem																						
L-19	Ford–Fulkerson, Edmonds–Karp algorithm																						
L-20	Convex hull problem, Graham’s scan, Jarvis march and quick hull (overview)																						
L-21	Backtracking approach, N-Queens, graph coloring, subset sum, Hamiltonian cycle	Other Advanced algorithms		CO1						10			10										
L-22	Branch and bound technique, knapsack, traveling salesman problem																						
L-23	Naïve String matching, KMP algorithm, Rabin–Karp (overview)																						
L-24	P and NP classes, NP-hard problems, decision vs optimization problems, NP-completeness	Complexity Classes: P, NP, and Hard Problems																5		5			
L-25	NP-complete problems (SAT, 3-CNF, clique), approximation algorithms																						
L-26	Review of the Topics																						
L-27	Final Exam																						
Mark Distribution (Theory Part)					15		25		30														

Laboratory Experiments and Assessment Scheme

Labs		Experiments	Teaching Learning Method	CO	Marks						
					Cognitive Learning Levels		Psychomotor Learning Levels		Affective Learning Levels	CO	PO
					C3	C4	P2	P3	A2	CO4	PO2
Lab 1	Lab Experiments	Brute-force approaches, complexity optimization, linear & binary search	Lab Experiment, Class Discussion, Group task & Presentation, Lab Manual and Report	CO4						10	25
Lab 2		Basic sorting algorithms (Insertion, Bubble, Selection)									
Lab 3		Divide-and-Conquer (Mergesort, Quicksort)									
Lab 4		Greedy algorithms (Min Coin Change, Fractional Knapsack, Activity Selection, Huffman Coding)									
Lab 5		Dynamic Programming (Memoization vs. Tabulation, 0/1 Knapsack, Coin Change)									
Lab 6		Other DP problems (LCS, LIS, Matrix Chain Multiplication) with practice in both memoization and tabulation techniques.									
Lab 7		Graph representation, MST algorithms (Prim’s, Kruskal’s)									
Lab 8		Shortest path algorithms (Dijkstra’s, Bellman–Ford, Floyd–Warshall)									
Lab 9		Network flow algorithms (Ford–Fulkerson, Edmonds–Karp)									
Lab 10		String matching Algorithms (Naïve, KMP, others)									
Lab 11		Selected geometric, backtracking, and branch-and-bound algorithms									
Lab 12	Lab Final									10	
Lab 13	Mini Project Showcasing									5	

Mini Project

Mini Project	Teaching-Learning Method	CO	EP/EA	Marks						
				Cognitive Learning Level		Psycho-motor Learning Level		Affective Learning Level	CO	PO
				C3	C4	P2	P3	A2	CO4	PO2
Mini Lab Project including Report and Presentation	Group based moderately complex project with report writing, and oral/poster presentation	CO4	EP2, EP3, EP4						5	5
Dates of report submission and presentation will be announced in the class.										

Overall Assessment Scheme

Assessment Area		CO Marks				PO Marks		Sub-Total
		CO1	CO2	CO3	CO4	PO1	PO2	
Class Participation	5							5
Class Test		5	5	5		10	5	15
Mid Exam		5	20			25		25
Final Exam		5		25		5	25	30
Lab Exam					10		10	10
Lab Performance					10		10	10
Mini Projects					5		5	5
Total Mark	5	15	25	30	25	40	55	100

Teaching Materials/Equipment

Referred Textbooks:

1. Introduction to Algorithms (4th edition) by T. H. Cormen, C. E. Leiserson, R. Rivest, C. Stein
2. Algorithms (4th edition) by Robert Sedgewick and Kevin Wayne

Grading Systems

Marks (%)	Letter Grade
80-100	A+
75-79	A-
70-74	A-
65-69	B+
60-64	B

Marks (%)	Letter Grade
55-59	B-
50-54	C+
45-49	C
40-44	D
0-39	F

Exam Dates

Can be found on the website.

Academic Code of Conduct

Academic Integrity

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter the classroom 20 minutes after the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the **mid-semester exam**, the student **MUST** get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application. **It is the responsibility of the student to arrange a Makeup Exam in consultation with the Course Instructor.**
- For **final exam**, there will be **NO** makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones **MUST** be turned into silent mode during class and exam period.
- There is **zero tolerance for cheating** in exams. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinees, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**